CS & IT ENGINEERING



Computer Networks

Application Layer Protocols

DPP 01 Discussion Notes



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[MSQ]

#Q. In the following pairs of OSI protocol layers and their functionality, which one is the CORRECT pair / s?

1. Network layer and Routing
2. Data Link Layer and Bit synchronisation
3. Transport layer and End-to-end process communication
4. Medium Access Control sublayer and Channel sharing

Physical Layer

Random
Controlled
Channel

A 1 & 2 only

B 2 & 3 only

C 1, 2 & 3

D 1, 3 & 4 only

[MSQ]

#Q. In the following pairs of OSI protocol layer and their functionality, which one is the CORRECT pair / s?

1. Segmentation and reassembly are done at hosts only, not at each intermediate router.
2. The Application Layer allows the user or application to access the network.
3. The Session Layer establishes, maintains, and synchronises the interaction between the communicating systems.
4. The Presentation Layer allows the process to add checkpoints or synchronisation points in the stream of data.

Syntax & Semantics

A. 1 & 2 only

B. 2 & 3 only

C. 1, 2 & 3

D. 3 & 4 only

[MSQ]

#Q. Which of the following is/are CORRECT?

1. Physical addresses in the packet are modified from HOP to HOP.
2. Network Layer ensures packets belonging to the same message arrive intact and in order.
3. The network layer uses physical addresses for addressing.
4. The Transport Layer treats every packet individually.

logical

A 1 only

B 2 & 3 only

C 1, 2 & 3

D 3 & 4 only

[MSQ]

#Q. Choose the correct statement/s:

1. The unit of communication at the physical layer is a bit.
2. The unit of communication at the network layer is a datagram.
3. The unit of communication at the transport layer is a segment.
4. The unit of communication at the transport layer is a user datagram or a packet

A 1 & 2 only

B 2 & 3 only

C 1, 2 & 3

D 1, 2, 3 & 4

[MSQ]

#Q. Choose the incorrect statement/s:

1. The computer used for forwarding IP packets is called an IP router
2. The Session Layer decides whether the way of communication between them is Half Duplex or Full Duplex.
3. The physical addresses will change from hop to hop, but the logical and port addresses usually remain the same
4. The two most typical network applications that use UDP are File Transfer Protocol (FTP) and the TELNET

Reliability

TCP

A. 1 & 2 only

B. 2 & 3 only

C. 1 & 3 only

D. 4 only

[MCQ]

#Q. Match the protocols from Group I to one or more in Group II:

Group I		Group II (port, TCP/UDP)	
1.	TELNET A	A.	23 TCP
2.	SMTP B	B.	25 TCP
3.	~~SMTP~~ DHCP-SERVER C	C.	67 UDP
4.	FTP - DATA D	D.	~~67 UDP~~ 20 TCP
5.	FTP - CONTROL E	E.	~~67 UDP~~ 21 TCP
6.	~~FTP - CONTROL~~ DHCP-CLIENT F	F.	68 UDP
		G.	67 TCP
		H.	~~67 TCP~~ 68 TCP

A. 1- A,2-B,3-C,4-D,5-E,6-F

B. 1- A,2-B,3-C,4-D,5-E,6-F

C. 1- A,2-B,3-C,4-D,5-E,6-F

D. 1- A,2-B,3-C,4-D,5-E,6-F

[MCQ]

#Q. Which layer of the ISO-OSI layer is responsible for the synchronisation of the stream of data?

A. Session Layer

B. Presentation Layer

C. Network Layer

D. Data Link Layer

THANK - YOU